

Software Test Plan

countdown: An X Window Countdown Timer

Countdown Software Project

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1 Introduction

1.1 Document identifier

CD-STP-001 Rev. 1.0, *Software Test Plan: countdown*.

1.2 Purpose

This Software Test Plan (STP) specifies how the requirements of CD-SRS-001 shall be verified by execution and static inspection. It identifies the test items, the test environment, the test cases and their pass/fail criteria, the requirement-to-test traceability, and the deliverables.

1.3 Scope

The STP covers the `countdown` executable tangled from `countdown.w`. Tests divide into two classes: *automated* cases, run headless by the harness `test-countdown.sh` (driven by `run-stp.sh`); and *manual* cases, performed by an operator at a live X display where automation would require a windowing server and human visual confirmation.

2 Test items

- `countdown.w` — the literate source of record.
- `countdown.c` — the tangled C source (build artefact).
- `countdown` — the compiled executable.
- `Makefile` — the build driver.

3 Features to be tested

All requirements in CD-SRS-001 §3 are in scope: functional (REQ-F-*), GUI (REQ-G-*), timing (REQ-T-*), design constraint (REQ-D-*), and build (REQ-B-*).

4 Features not to be tested

The correctness of the X server, the host's time-zone database, the C compiler, and the CWEB tools themselves are outside scope; they are assumed correct.

5 Test environment

Automated. A POSIX host with `cc`, `make`, a Python 3 interpreter, and the CWEB tools (`ctangle/cweave`, reached through `cweb.sh`). No X server is required: every automated case either inspects source or exercises the program's pre-display code paths.

Manual. Additionally an X display (Linux/X11 or macOS/XQuartz), at which an operator can observe the window and operate the keyboard and mouse.

6 Approach

Automated cases emit one line per requirement of the form **PASS/FAIL/SKIP** followed by the requirement identifier and a short description. A case is **SKIPPED** only when its precondition (e.g. a live X display) is absent. The run exits with a non-zero status if any case fails. Manual cases are recorded on the checklist in Section 8.

7 Automated test cases

Case	Req.	Procedure / pass criterion
TC-01	REQ-B-01, REQ-D-04	<code>make tangle</code> regenerates <code>countdown.c</code> from <code>countdown.w</code> ; the executable builds.
TC-02	REQ-D-05	The compile produces no warning : lines under <code>-Wall</code> .
TC-03	REQ-B-02	The link line references <code>-lX11</code> and no widget toolkit.
TC-04	REQ-D-01	<code>countdown.w</code> contains CWEB markers (<code>@*</code> , <code>@<...@>=</code>) and names literate programming.
TC-05	REQ-D-02	No CWEB code section exceeds 23 lines of code (checked by script).
TC-06	REQ-D-03	A “POSIX system services” section exists and the index records each system call (<code>time</code> , <code>mktime</code> , <code>localtime</code> , <code>select</code>).
TC-07	REQ-F-02	The <code>DEFAULT_TARGET</code> macro equals <code>2026-08-19 08:00:00</code> .
TC-08	REQ-F-03	The source accepts <code>argv[1]</code> as the target specification.
TC-09	REQ-F-04	Running <code>countdown "not-a-date"</code> prints a diagnostic and exits with status 2, opening no window.
TC-10	REQ-F-05	<code>fmt_duration</code> emits the <code>d HH:MM:SS</code> form with zero-padding.
TC-11	REQ-F-06	<code>remaining_secs</code> clamps a past target to zero (verified via a past-dated target with status 2 guard / source inspection of the clamp).
TC-12	REQ-T-02, REQ-T-03	The source computes remaining time with <code>time</code> and converts the target with <code>mktime</code> using <code>tm_isdst = -1</code> .
TC-13	REQ-T-04	The event loop blocks on <code>select</code> (no busy-wait).

8 Manual test cases

Case	Req.	Procedure / pass criterion
MC-01	REQ-G-01, REQ-G-02	Launch <code>countdown</code> ; a window opens showing target, remaining time, status, and hint.

Case	Req.	Procedure / pass criterion
MC-02	REQ-F-01, REQ-T-01	With no argument, the window counts down toward the default target, the seconds advancing once per second.
MC-03	REQ-G-03	Press space (or click): the status toggles running↔paused and the figure freezes when paused.
MC-04	REQ-G-04	Press q or Escape: the window closes and the program exits with status 0.
MC-05	REQ-G-05	Resize and re-expose the window: contents are repainted correctly.
MC-06	REQ-F-07	Launch with a target a few seconds ahead; on arrival the status shows the arrival message and the figure reads 0 d 00:00:00.

9 Pass/fail criteria

The test campaign passes when every automated case reports **PASS** (or a justified **SKIP** for an absent precondition) and every manual case is confirmed by the operator. Any **FAIL** fails the campaign and blocks release.

10 Traceability matrix

Requirement	Verified by	Method
REQ-F-01	MC-02	Manual
REQ-F-02	TC-07	Automated (static)
REQ-F-03	TC-08	Automated (static)
REQ-F-04	TC-09	Automated (dynamic)
REQ-F-05	TC-10	Automated (static)
REQ-F-06	TC-11	Automated (static)
REQ-F-07	MC-06	Manual
REQ-G-01	TC-03, MC-01	Automated + Manual
REQ-G-02	MC-01	Manual
REQ-G-03	MC-03	Manual
REQ-G-04	MC-04	Manual
REQ-G-05	MC-05	Manual
REQ-G-06	TC-04 (fallback code present)	Automated (static)
REQ-T-01	MC-02	Manual
REQ-T-02	TC-12	Automated (static)
REQ-T-03	TC-12	Automated (static)
REQ-T-04	TC-13	Automated (static)
REQ-D-01	TC-04	Automated (static)

REQ-D-02	TC-05	Automated (static)
REQ-D-03	TC-06	Automated (static)
REQ-D-04	TC-01	Automated (dynamic)
REQ-D-05	TC-02	Automated (dynamic)
REQ-B-01	TC-01	Automated (dynamic)
REQ-B-02	TC-03	Automated (static)

11 Test deliverables

- This plan (CD-STP-001).
- The automated harness `test-countdown.sh`.
- The plan runner `run-stp.sh`.
- The harness console log and exit status from each run.

12 Responsibilities and schedule

The maintainer runs the automated harness before every release; a 0 exit status is a precondition for tagging. The manual checklist (Section 8) is performed at a live display whenever GUI-affecting code changes.